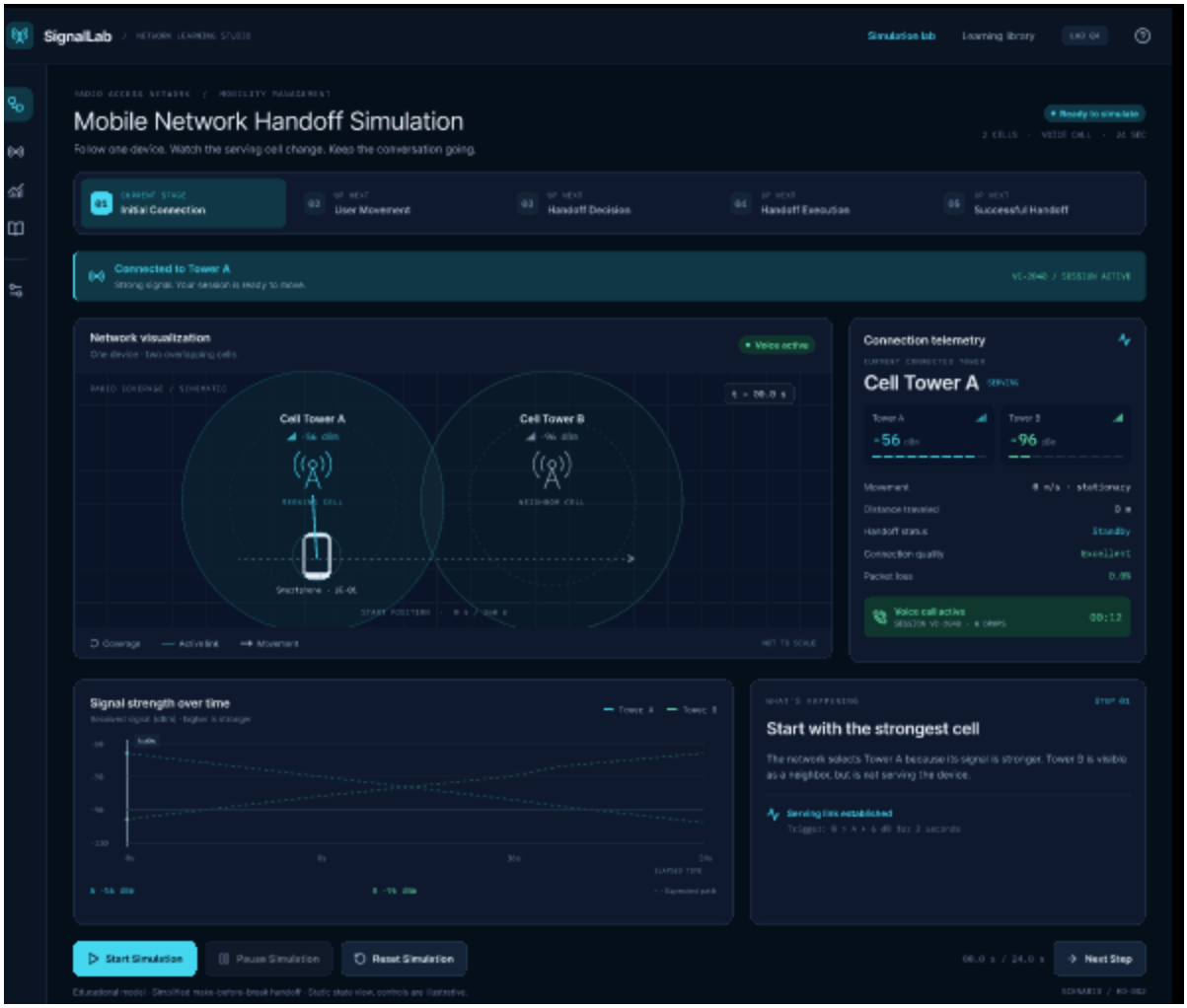
6. Use Figma to simulate a handoff scenario in a mobile network and visualize the process

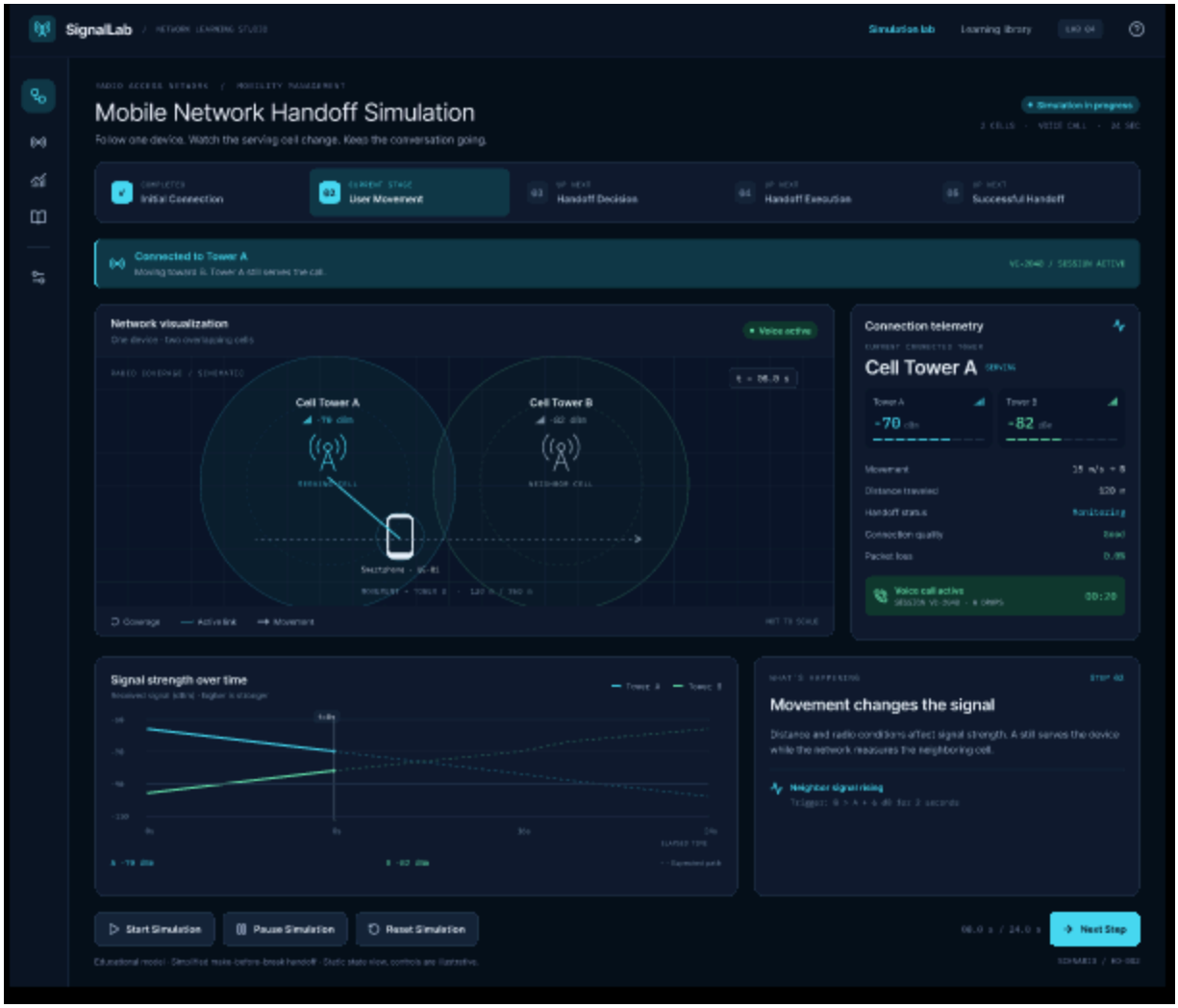
Aim: -To simulate a handoff scenario in a mobile network and visualize the process using Figma.

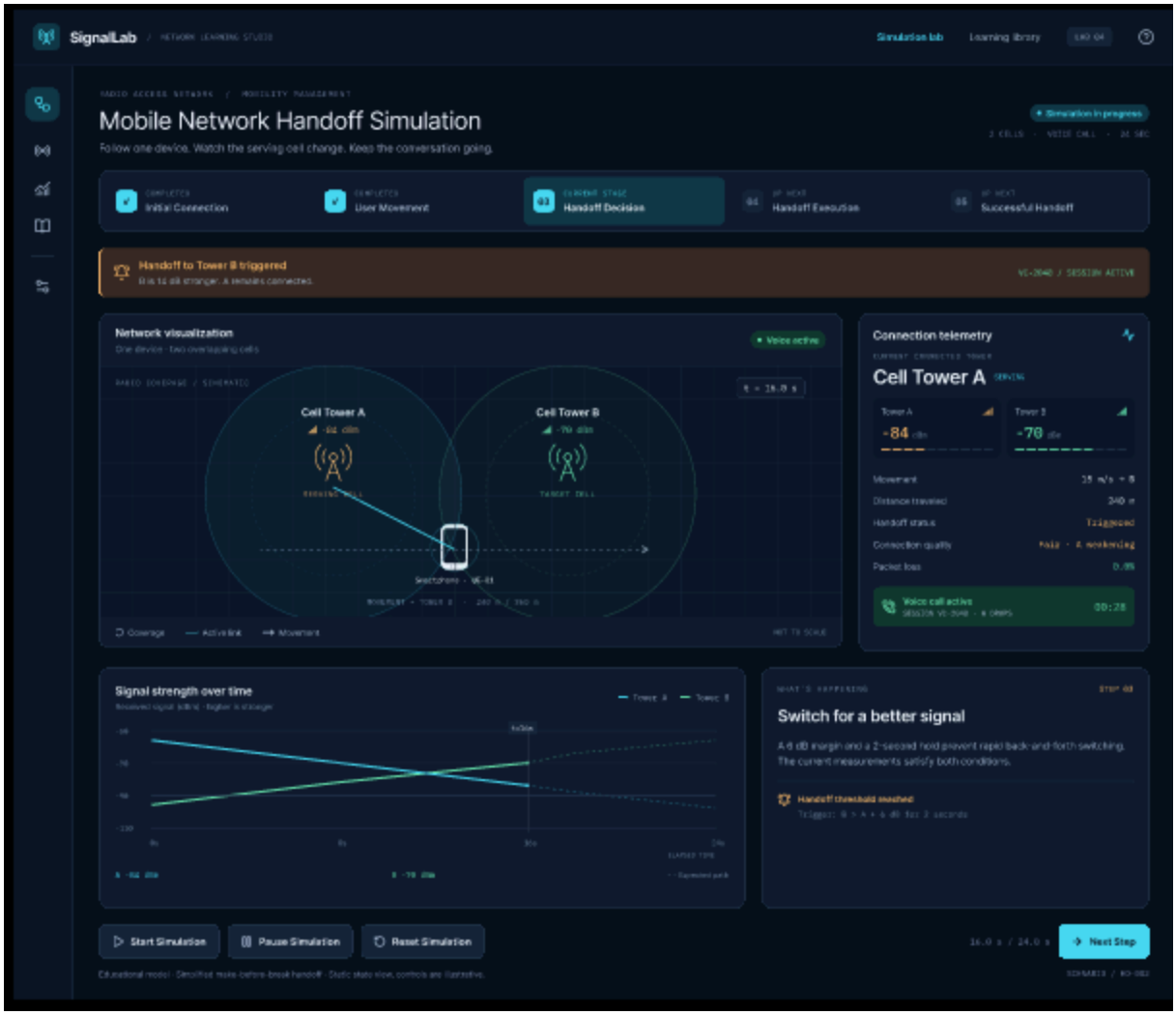
Procedure:

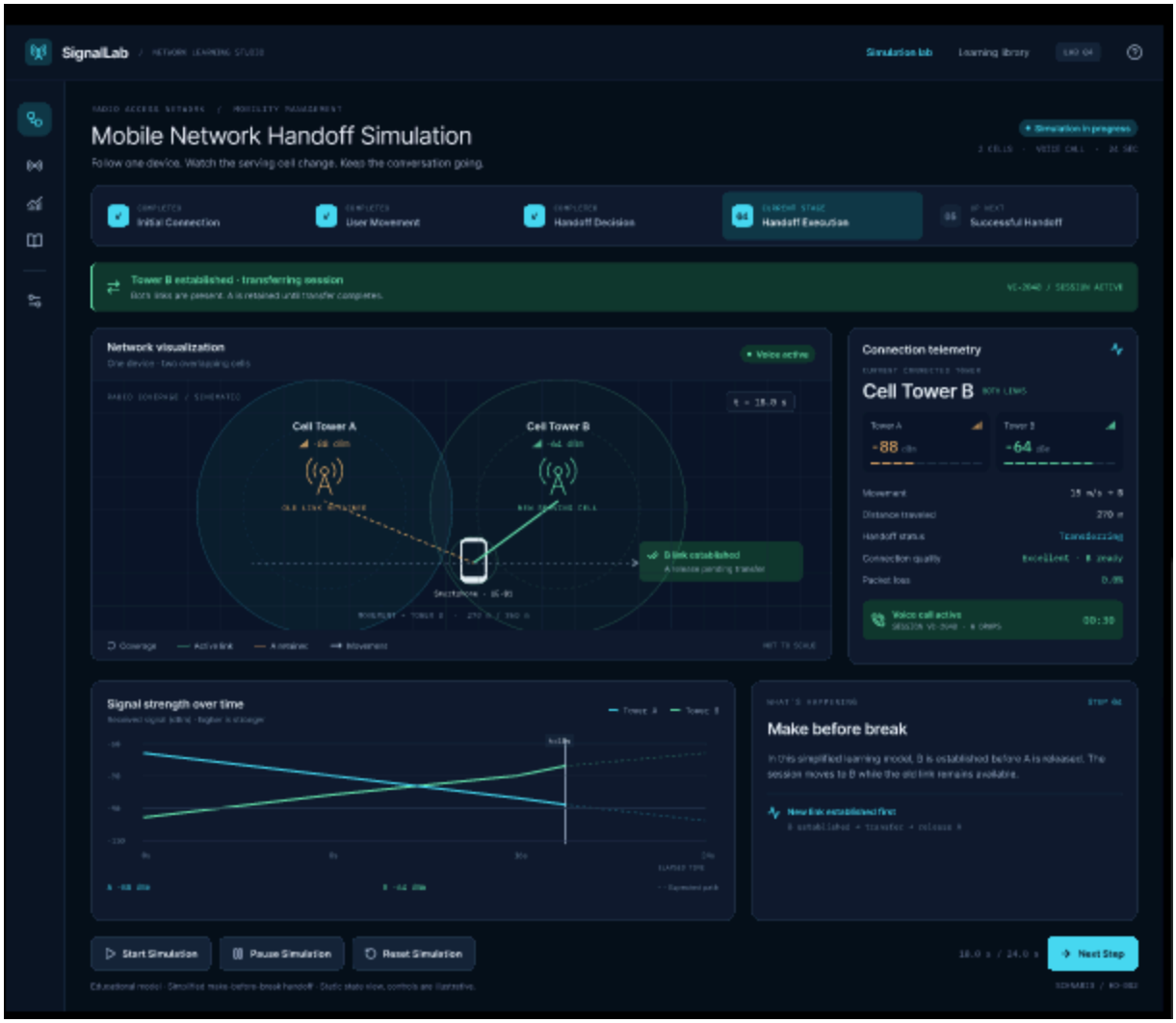
1. Open Figma
2. Create a new file
3. Select the Frames
4. Fill in the content that is required for presentation
5. Design Visual Elements
6. Make it Interactive
7. Add Annotations and Explanations
8. Incorporate Multimedia
9. Storyboard Animation
10. Review and edit the Prototype
11. Save and Share

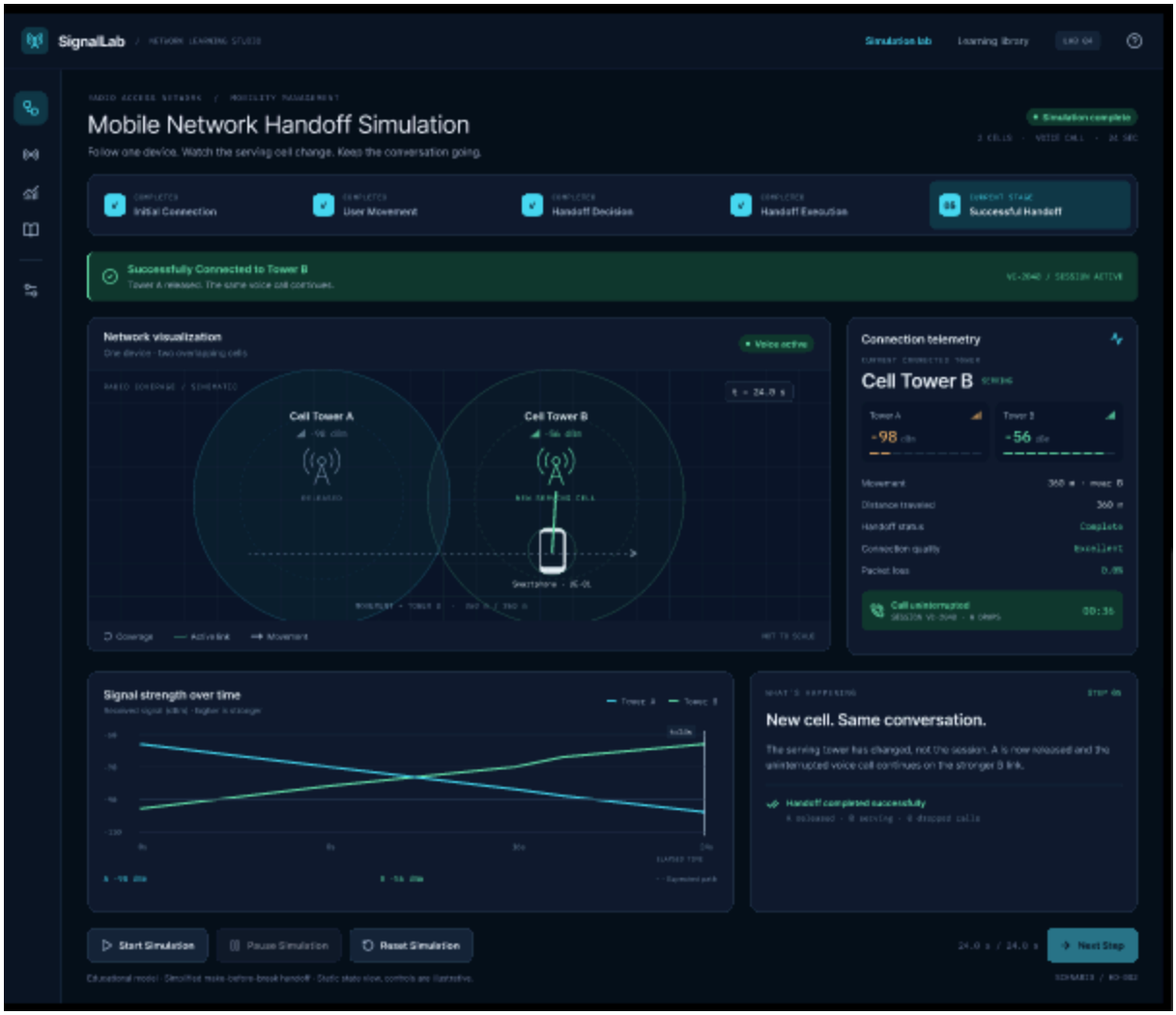
Design:


SignalLab
Simulation lab
Learning library
Mobile Network Handoff Simulation
Follow one device. Watch the serving cell change. Keep the conversation going.
Ready to simulate
01 Initial Connection
02 User Movement
03 Handoff Decision
04 Handoff Execution
05 Successful Handoff
Connected to Tower A
Strong signal. Your session is ready to move.
Network visualization
One device · two overlapping cells
Voice active
Cell Tower A
Cell Tower B
Smartphone · UE-01
Coverage
Active link
Movement
Connection telemetry
Cell Tower A
Serving
Tower A
-56 dBm
Tower B
-96 dBm
Movement
0 m/s · stationary
Distance traveled
0 m
Handoff status
Standby
Connection quality
Excellent
Packet loss
0.0%
Voice call active
00:12
Signal strength over time
Tower A
Tower B
Start with the strongest cell
The network selects Tower A because its signal is stronger. Tower B is visible as a neighbor, but is not serving the device.
Serving link established
Start Simulation
Pause Simulation
Reset Simulation
Next Step


SignalLab
Simulation lab
Learning library
Mobile Network Handoff Simulation
Follow one device. Watch the serving cell change. Keep the conversation going.
Simulation in progress
Initial Connection
User Movement
Handoff Decision
Handoff Execution
Successful Handoff
Connected to Tower A
Network visualization
Voice active
Cell Tower A
Cell Tower B
Connection telemetry
Cell Tower A
-70
-82
Movement
Distance traveled
Handoff status
Connection quality
Packet loss
Voice call active
Signal strength over time
Movement changes the signal
Distance and radio conditions affect signal strength. A still serves the device while the network measures the neighboring cell.
Start Simulation
Pause Simulation
Reset Simulation
Next Step


SignalLab
Simulation lab
Learning library
Mobile Network Handoff Simulation
Follow one device. Watch the serving cell change. Keep the conversation going.
Simulation in progress
Initial Connection
User Movement
Handoff Decision
Handoff Execution
Successful Handoff
Handoff to Tower B triggered
Network visualization
Voice active
Cell Tower A
Cell Tower B
Connection telemetry
Cell Tower A
Tower A
-84
Tower B
-76
Movement
Distance traveled
Handoff status
Triggered
Connection quality
Packet loss
Voice call active
00:28
Signal strength over time
Switch for a better signal
A 6 dB margin and a 2-second hold prevent rapid back-and-forth switching. The current measurements satisfy both conditions.
Handoff threshold reached
Start Simulation
Pause Simulation
Reset Simulation
Next Step


SignalLab
Simulation lab
Learning library
Mobile Network Handoff Simulation
Follow one device. Watch the serving cell change. Keep the conversation going.
Simulation in progress
Initial Connection
User Movement
Handoff Decision
Handoff Execution
Successful Handoff
Tower B established - transferring session
Network visualization
One device · two overlapping cells
Voice active
Cell Tower A
Cell Tower B
B link established
A release pending transfer
Coverage
Active link
Movement
Connection telemetry
Cell Tower B
Tower A
-88 dBm
Tower B
-64 dBm
Movement
Distance traveled
Handoff status
Connection quality
Packet loss
Voice call active
Signal strength over time
Make before break
In this simplified learning model, B is established before A is released. The session moves to B while the old link remains available.
Start Simulation
Pause Simulation
Reset Simulation
Next Step



Result: Hence the simulated a handoff scenario in a mobile network and visualized the process using Figma.